

Assignment 2 Report

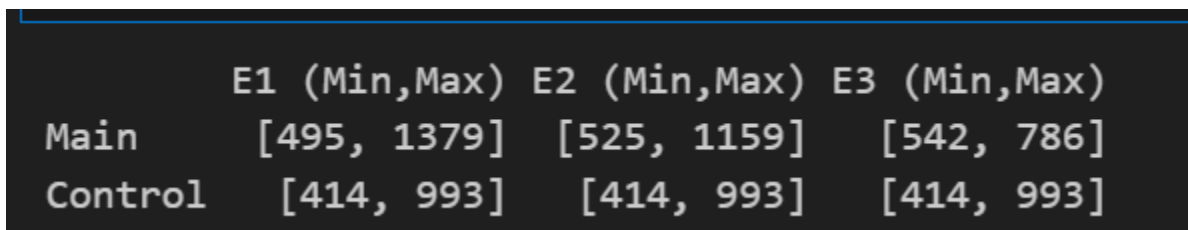
Introduction

This research examines the relation between pupil size (pupil dilation) when a person perceives various emotional facial features. In particular, we are interested in how participants deal with three emotions- Happy, Angry and Neutral, when carrying out an emotion recognition task (main experiment) and a control experiment where participants simply have to find the location of a red dot on a face.

In the main experiment, there are participants who display emotions so the subsequent changes in pupil dilation and latency is observed..

Data inspection

The “copy of target.csv” file is similar across both Main and Control. The first column contains the name of the face stimulus presented. The second column represents the time (in milliseconds) the participant took the time to respond after the stimulus presented. The third column indicates the key pressed by the participant to register their response (e.g., 'Left' or 'Right'). And the subsequent columns contain pupil diameter measurements taken at 8 ms intervals throughout the trial. I also reviewed the minimum and maximum number of pupil diameter columns across all E1, E2, and E3 blocks for both Main and Control experiments to understand the data's variability. The figure is attached below.



	E1 (Min,Max)	E2 (Min,Max)	E3 (Min,Max)
Main	[495, 1379]	[525, 1159]	[542, 786]
Control	[414, 993]	[414, 993]	[414, 993]

Figure 1: Minimum and maximum number of columns across each experiment and each emotions

Also, I explored that there is a mismatch in the number of samples (i.e., the number of trials or rows) across the E1, E2, and E3 files in both Main and Control experiments. Figure 2 below, showing the variation in sample sizes between the emotion blocks for each experiment.

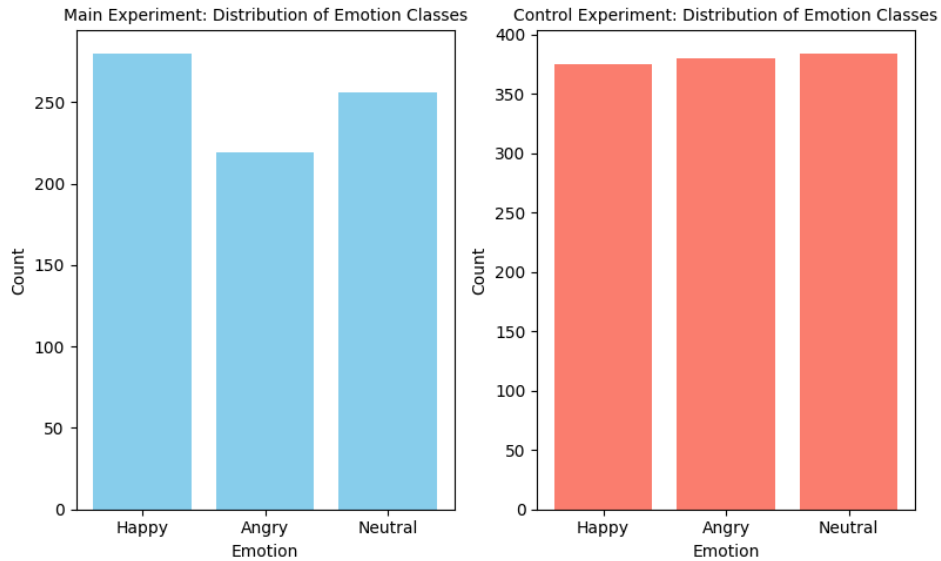


Figure 2: Number of samples across all emotions

These variations in the sample size across all emotions may affect the calculation and through this plot I infer that there is a slight imbalance in the dataset.

Preprocessing

I first checked whether the "Copy of target.csv" files are present in the E1, E2, and E3 folders within the "Main" and "Control" directories for 10 individuals. If any file was missing, I printed the respective path. Then, for each existing file, I examined if there were any columns containing only NaN values, and removed those columns. Additionally, I dropped the column preceding t87, as it contains pre-stimulus pupil dilation data. And also drop the last column because it has no significance in the both experiment. Then the cleaned data was then saved in a new CSV file which is as "new_file.csv".

After preprocessing the data, I have re-evaluated the minimum and maximum number of pupil diameter columns across E1, E2, and E3 blocks for both Main and Control experiments. This revealed a reduction in the number of columns due to the removal of NaN-only columns. The following are the number of columns:

	E1 (Min,Max)	E2 (Min,Max)	E3 (Min,Max)
Main	[341, 825]	[364, 534]	[365, 696]
Control	[324, 723]	[312, 804]	[318, 903]

Figure 3: Updated minimum and maximum number of columns across each experiment and each emotions

Data Analysis

Pupil diameter

This section provides an overview of how pupil dilation and reaction times change in response to different emotional stimuli in both Main and Control tasks.

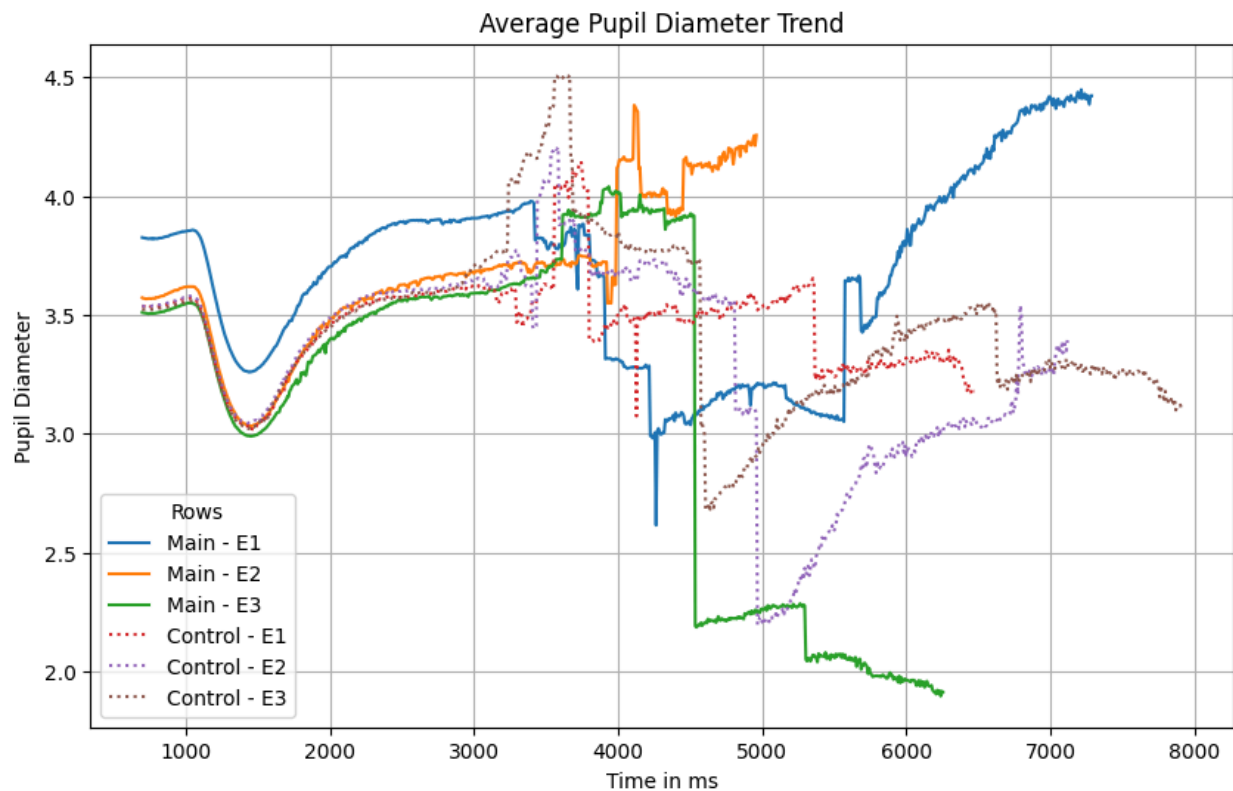


Figure 4: Pupil dilation across all emotions

The graph begins around 689 ms with regard to the pupil diameter because it contains pre-stimulus pupil dilation time. E1, E2 and E3 from Main show noticeable increase from those of control experiments. All the Es start decreasing from the 1024ms the pupil diameter reduction begins over all of the performed experiments combined with emotions. By approximately 1422 ms the pupil diameter is seen at its lowest level for Main of E3.

Reaction time

In addition to examining pupil dilation, the study also analyzed reaction times across both the main and control experiment to determine if emotional processing affected participants' response speeds. The following steps were taken to process reaction time data.

We calculated the median reaction time for each emotion (Happy, Angry, Neutral) in the main and control experiment. Medians were selected over means to reduce the impact of any remaining outliers on the final analysis.

The figure 5 illustrates the reaction time for the main and control experiment.

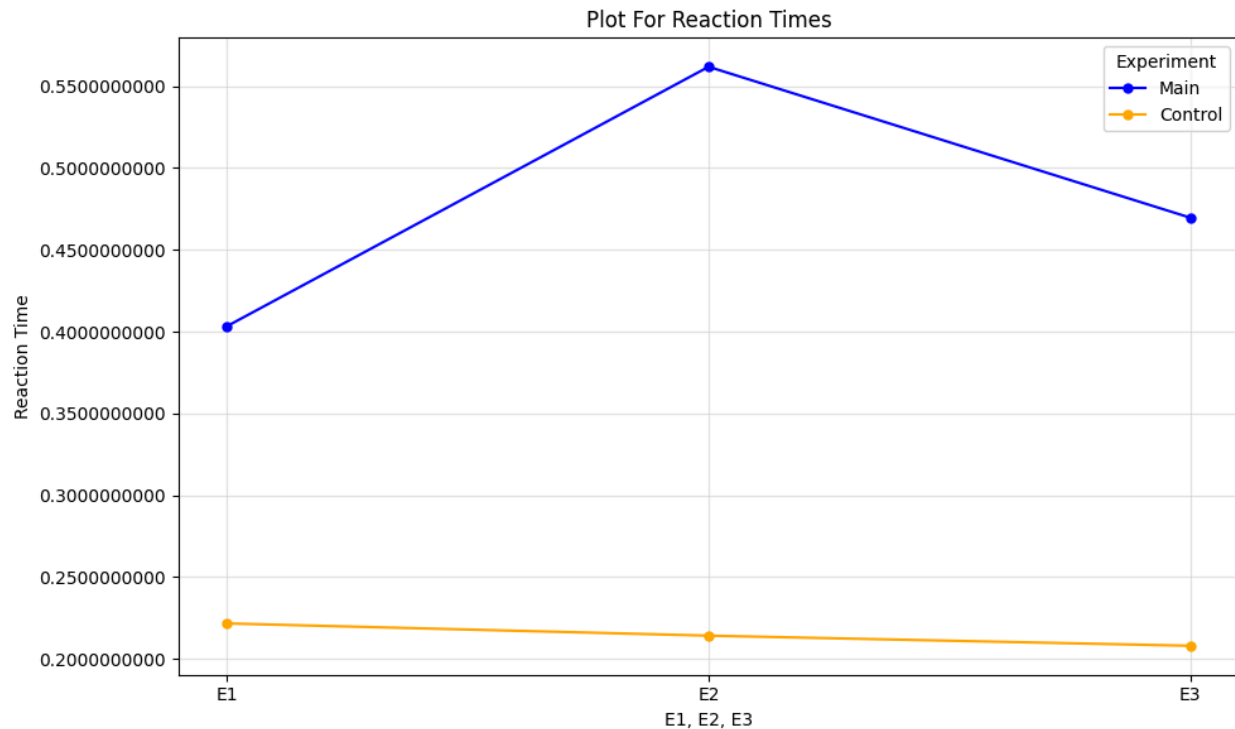


Figure 5: Shows the median reaction times for each emotion in both the main and control tasks.

The graph shows the RTs across 3 conditions (E1, E2, E3) in two experiments, the Main (M, blue line) and the Control (C, orange line). As demonstrated in the E1, E2 and E3 comparison results from the Main experiment, reaction time is maximum in E2 of about 0.567 but minimum in E1 and E3. This implies that different cognitive processes are used in the two conditions.

For all the control experiments E1 and E2 and E3, the median reaction times are quite stable and low. This seems to indicate that the main experiment is more difficult for an

emotional response. The Control experiment appears to be stable due to uncomplicated tasks.

Standard Deviation of Pupil dilation

Moreover, I have shown the standard deviation of pupil dilation, which is calculated for each block (E1, E2, E3) in both main and control experiments to gauge the variability in physiological responses. This analysis provides insight into whether the main task induced greater variability, potentially due to emotional processing demands.

Figure 6 illustrates the standard deviation of pupil diameter for each emotion block in both the main and control tasks.

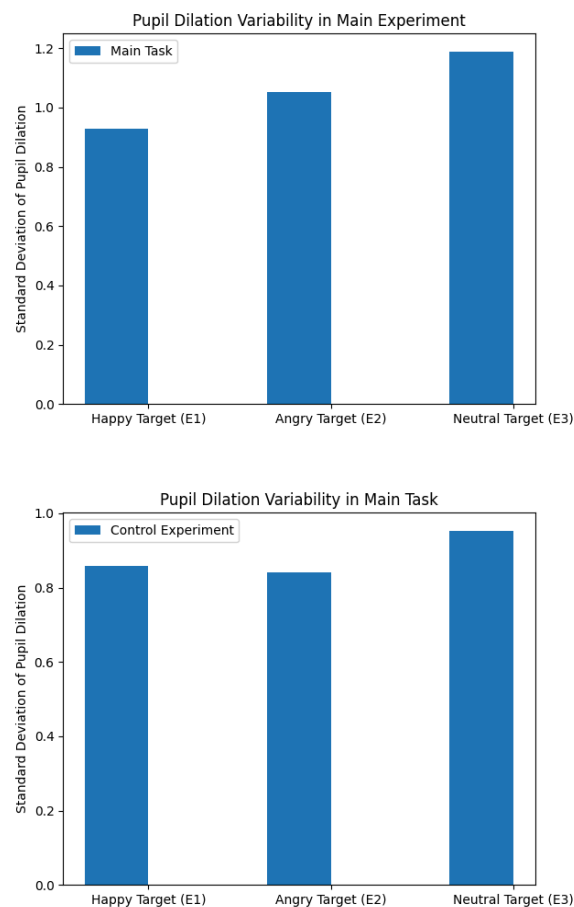


Figure 6: standard deviation of pupil dilation

The main task displayed higher variability in pupil responses across all emotions, suggesting greater cognitive engagement and arousal in emotion recognition. This variability supports the idea that the emotional processing in the main task

was more cognitively demanding, resulting in wider fluctuations in pupil dilation compared to the control Experiment.

Inference

There is a substantial change with regard to pupil diameter. In the Main experiment angry faces lead to the most pupil dilation as compared to happy or neutral faces, suggesting that negative emotions tend to cause a significantly higher physiological arousal.

In the Main experiment the pupil response is exaggerated, especially with regard to the Angry faces because of the emotional recognition task. On the other hand, the other experiment shows a more uniform pupil diameter regardless of the emotion, possibly because participants are required to only fixate at the dot. This fixation of the dot has less cognitive load on the participants as compared to identifying the reaction.

In the Main experiment, reaction times are not consistent: for instance, Happy faces (E1) take the longest response time, Angry faces (E2) have the quickest response, while neutral faces (E3) take the intermediate times. This would indicate that the participants are quicker to respond to images of angry faces.

In the Main experiment, Anger faces are targeted since it is expected that they will bring out the strongest, fastest, and most varied reaction in terms of pupil response. which suggests that negative emotions are more physiologically arousing. In the Control experiment however, emotion, pupils and reaction times do not seem to correlate. This means that the cognitive component in comparison with the emotional one is considerably less complex.

The greater degree of change in pupil size for the main experiment as opposed to the control task implies that emotional processes such as recognizing facial expressions are more demanding in terms of physical arousal than a simple position dot task.

Conclusion

The study reveals that pupil dilation and reaction times are more pronounced in response to emotional stimuli, particularly for angry faces, in the main experiment compared to the control task. The heightened pupil responses indicate stronger physiological exhilaration when recognizing emotions, especially negative ones. This suggests that emotion recognition tasks impose a higher cognitive load than simpler

visual tasks. Overall, emotional processing requires greater cognitive and physiological engagement.

References

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